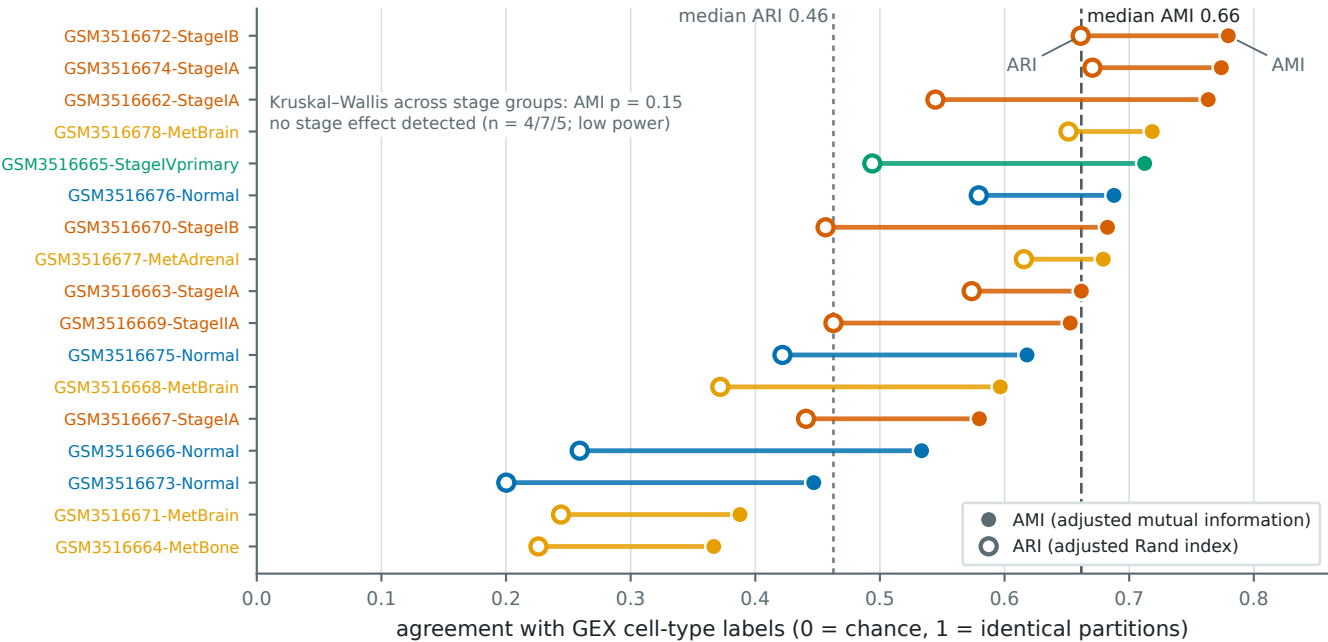


Clustering on poly(A)-site profiles alone recovers gene-expression cell-type structure

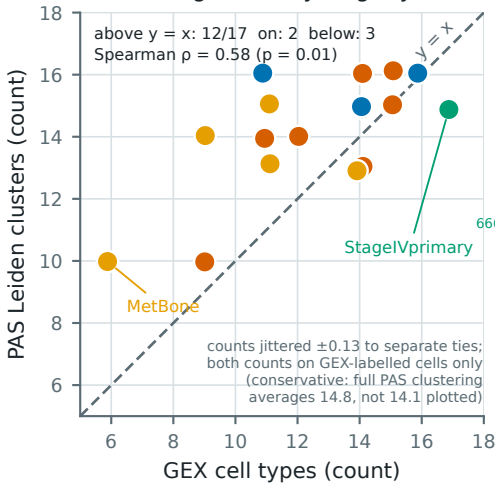
17 Laughney lung-cancer samples · PAS clusters use no gene-level expression matrix · colour = clinical stage group
Reference partition = marker-signature labels assigned per GEX Leiden cluster — an automated partition, not curated ground truth.
Features are per-site counts (TF-IDF + LSI), which carry gene abundance — this figure does not isolate isoform-usage signal.

● Normal (n=4) ● Stage I-II (n=7) ● Stage IV primary (n=1) ● Metastasis (n=5)

a PAS-only clusters recover GEX cell types in all 17 samples (median AMI 0.66, ARI 0.46)



b PAS space finds structure of the same granularity, slightly finer



c Caveat: concordance is scored on a minority of PAS cells in 9 of 17 samples

